

AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

53

Automotive Front/Rear Bumper Fascia

1691 × 647 × 528 mm · 2059.9 cm³ · AI 5.562 kg / Steel 16.170 kg

Manufacturability: 53/100 · Confidence: Low

§1 — Geometry & Part Summary

Bounding Box	1690.9 × 647.4 × 528.1 mm	Surface Area	16261.7 cm ²
Volume	2059.90 cm ³	Weight (AI)	5.562 kg
Weight (Steel)	16.170 kg	Weight (Plastic)	2.163 kg

§2 — Detected Features

Feature Type	Count	Significance	Description
Free-form B-spline surfaces	492	High	492 of 498 faces are B-spline — organic styled exterior skin typical of a bumper fascia
Undercuts	4	High	4 undercut faces requiring side-action cores or lifters to release from a single-direction draw
Zero-draft faces	2	Medium	2 faces with no draft angle — will gall/scuff on ejection
Thread-like feature flagged	1	Low	Detected 'threaded feature' likely a mounting boss/insert location rather than a true machine thread on a plastic part
Non-uniform thin wall	4	Medium	Wall varies 2.53–3.54 mm (∅ large due to shell-cast overshoot thickness transitions)

§3 — Material Analysis

AI-suggested

Primary Material:

PP Impact Copolymer (PP-B) (82% confidence)

Part geometry (large single-shell, organic double-curvature, ~2.5-3.5mm nominal wall, huge projected area, automotive bumper form factor) is the textbook application for PP impact copolymer — the dominant resin for automotive bumper fascias due to impact toughness, low cost at high volume, and paintability. Selected per user-forced injection_moulding process and part identity; no aluminium/steel signal applies since this is confirmed plastic (fill ratio 0.0036, hollow shell).

Alternatives:

PC/ABS Blend (automotive grade) (12%) · ABS (6%)

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Injection Moulding	injection_moulding	95%	0.0050	User-forced ground truth; geometry (thin uniform-ish shell, high B-spline surface content, hollow enclosed volume, huge projected area) is fully consistent with a single-cavity IM bumper fascia tool.

§5 — Manufacturability Risks (Score: 53/100)

Severity	Feature / Area	Description	Recommended Action
High	4 Undercuts	Undercut faces cannot release with a straight +Z pull; will require side-action cores/lifters, adding tool complexity and cost	Redesign undercut features to be moulded with lifters, or split as a separate clip-on component
High	Non-uniform wall (2.53–3.54mm)	Thickness transitions between 2.53mm and 3.54mm create differential cooling/shrinkage — high risk of sink marks and visible warp on the large flat/curved panel areas	Target a uniform 2.5–2.8mm nominal wall with smooth transitions (<15% thickness change) and add ribbing instead of thickening bosses
Medium	Zero-draft faces (x2)	Faces with no draft will drag/scuff on ejection, causing surface defects on a cosmetic (Class A) exterior surface	Add minimum 1–2° draft to all vertical/near-vertical faces, especially in the visible cosmetic zone
Medium	Large projected area / weld lines	10,946.8cm² projected area on a single-cavity tool will need multiple gates to fill in reasonable time, generating weld lines across the visible face, especially around logo/grille cut-outs	Use flow simulation to position gates away from Class-A visible zones and use sequential valve gating
Low	Threaded/insert feature	Detected thread-like feature is likely a moulded-in insert boss; if a true fine thread is required it cannot be moulded directly and needs a post-mould insert	Confirm feature is a boss for a brass insert, not a direct moulded thread

§6 — DFM Issues (injection_moulding)

Severity	Area	Description	Impact	Fix
High	Undercut/side-action zones	4 detected undercut faces will require lifters or side-action slides in the tool since a single +Z draw cannot release them		

§7 — AI Indicative Cost Range (model opinion — not engine-calculated)

Net Weight	1.854 kg	Material	mat-pp
Recommended Process	injection_moulding	Cycle Time	0.0050 hr/part
Setup Time	0.750 hr	Operations	0 ops

Operations Detail

Process-Specific Parameters

Casting Subtype	hpdc
Die/Mould Cost	£300,000
Die Life	100,000 shots
Cavities	1
Yield	85.0%
Flash Weight	0.000 kg
Yield	90.0%
Die Cost	£180,000
Strokes	7
Cavities	1
Wall Thickness	2.5 mm
Mould Cost	£312,157
Mould Life	1,000,000 shots
Projected Area	10946.8 cm²

§8 — AI Analysis Explanation